RAKSHAK (रक्षक / ૨ક્ષક)

Real-Time AI Guardian Against “Digital Arrest”, Fake KYC & Courier Scams

Team Rakshak
Powered by Google Gemini End-to-End Audio & Vertex AI

Indian Cyber Safety Initiative

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The Crisis: Indian Elders Under “Digital Arrest”

The Anatomy of the Scam

- **High-Stakes Extortion**: Scammers pose as Mumbai Police, CBI, ED, or FedEx Narcotics wing.
- **Isolation Weapon**: Victims are commanded into a closed room on a continuous video call (“Digital Arrest”).
- **Target Group**: Elders, retirees, and first-time smartphone users who are alone and vulnerable.
- **Scale**: Over **thousands of crores** looted across India annually.

Why Victims Comply

- Fear of arrest warrants & forged Supreme Court summons.
- Artificial urgency: “Transfer to RBI verification account in 15 mins”.
- **The Core Vulnerability**: The victim is **completely alone on the call** with no one to verify facts.

The Solution: Rakshak Silent Co-Pilot

How Rakshak Protects in Real-Time

Rakshak runs as a silent, on-device / edge guardian listening to incoming phone audio in **5-second sliding chunks**.

1. Stream Ingestion

Processes raw 16kHz PCM audio without transcribing to text first; preserves tone, hesitation, and pressure cues.

2. Mid-Call Alert

Flags scams at **15s–25s**, intervening **before** the victim hands over bank accounts or OTPs.

3. Vernacular Voice

Explains the scam in simple, reassuring **Hindi & Gujarati**: "Police never arrest on video call".

Why Gemini End-to-End Audio?

Capability	Browser Web Speech API	Rakshak (Gemini Audio)
Colloquial Dialects	Fails on Gujarati/Hindi code-switching	Native comprehension of Gujarati, Hinglish & regional nuances
Acoustic Robustness	Severely degrades under phone line noise & hiss	Trained on raw audio representations; ignores GSM compression
Context Window	Stateless per phrase	Accumulates session state across all prior chunks
Urgency & Threat Intonation	Strips vocal emotion	Detects aggressive caller tone & psychological pressure

Precision Scam Taxonomy

Decisive Red Flags

- **authority_impersonation**: Police, CBI, TRAI, RBI, or FedEx customs officials.
- **threat_or_arrest**: Threat of FIR, jail warrant, pension freeze.
- **secrecy_isolation**: “Do not disconnect”, “Do not tell your family”.
- **money_transfer**: Safe RBI account, clearance fee, UPI transfer.
- **otp_request**: Demanding OTP, PIN, or CVV.

Decision Logic

- Single high-confidence flag triggers **suspicious** status.
- Dual category correlation (e.g. Authority Impersonation + Arrest Threat) triggers **immediate mid-call intervention**.
- Skips follow-up LLM summarization for sub-second latency.

Live Demo Verification (Mid-Call Detection)

Timeline of Scripted Digital Arrest Call (`digital_arrest_hi.wav`)

- **0.0s – 5.0s** [SAFE]: “Caller asks for Ramesh Patel, confirms identity.”
- **5.0s – 10.0s** [SUSPICIOUS]: “Claims to be Inspector Vikram Singh, Mumbai Cyber Crime.”
- **15.0s – 25.0s** **[SCAM ALERT TRIGGERED (100% Confidence)]**

Generated Vernacular Warnings

हिंदी: कोई भी असली पुलिस अधिकारी फोन पर गिरफ्तारी की धमकी नहीं देता। फोन तुरंत काट दीजिए।
ગુજરાતી: કોઈપણ અસલી પોલીસ અધિકારી ફોન પર ધરપકડની ધમકી નથી આપતા. ફોન તરત જ કાપી નાખો.

Outcome

Alert raised **35 seconds before** the scammer asked for 4 Lakh rupees!

Guarantee: Never Cry Wolf

Genuine Call (`genuine_gu.wav`)

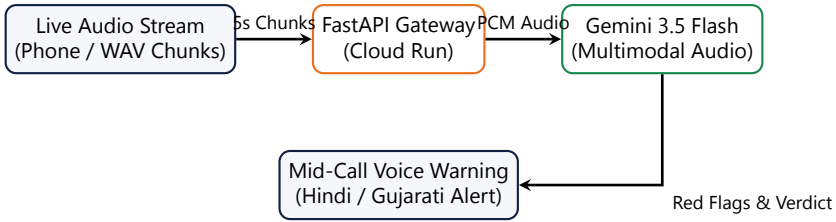
Context: Son calling his mother in Gujarati regarding dinner and doctor's appointment.

- **0–5s**: Greeted 'mummy', answered 'Ha mummy'.
- **5–10s**: Discussing health & medicine.
- **10–15s**: Doctor appointment at 10 AM.
- **Verdict**: **SAFE — 0 alerts raised**.

Contextual Safety Filters

- Delivery callers asking for society gate directions → **SAFE**
- Bank automated balances or legit OTP reminder → **SAFE**
- Advice like "never share OTP" → Recognized as a **protective positive sign**.

Cloud Native Architecture on Google Cloud



- **Google Cloud Run**: Serverless auto-scaling container backend.
- **Vertex AI Global**: Low-latency multimodal audio understanding.
- **A2A Protocol Ready**: Autonomous Agent interoperability.

Summary & Social Impact

Impact for Indian Citizens

- **Protection for the Most Vulnerable**: Replaces solitude with an intelligent guardian.
- **Linguistic Inclusivity**: Native support for India's regional languages.
- **Actionable Clarity**: Eliminates panic by debunking the lie before money is lost.

Rakshak: Defending Indian Families from Scams

Questions & Demonstration